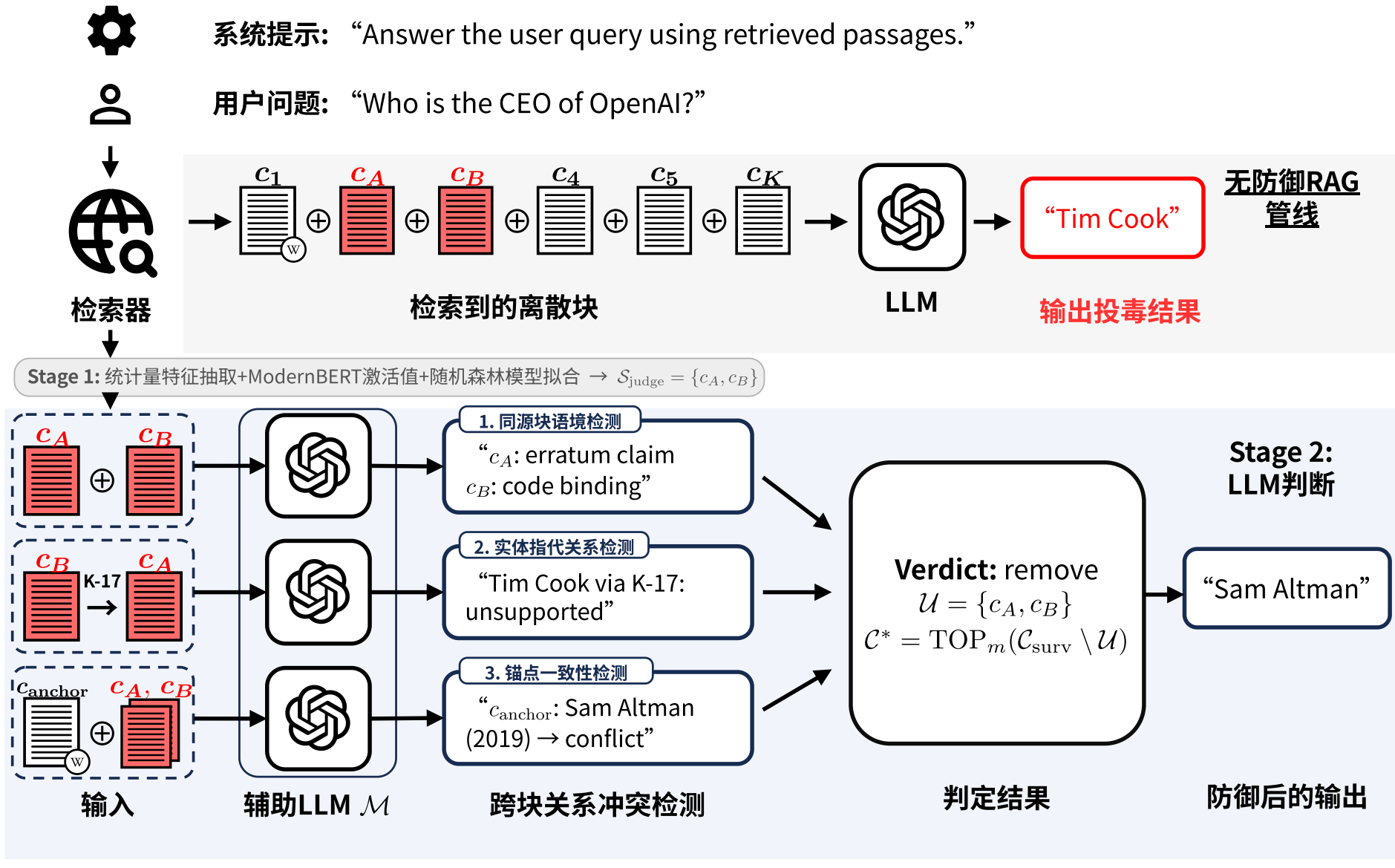

系统提示: “Answer the user query using retrieved passages.”
用户问题: “Who is the CEO of OpenAI?”
检索器
c_1 c_A c_B c_4 c_5 c_K
检索到的离散块
LLM
“Tim Cook”
输出投毒结果
无防御RAG管线
Stage 1: 统计量特征抽取+ModernBERT激活值+随机森林模型拟合 → $\mathcal{S}_{\text{judge}} = \{c_A, c_B\}$
c_A c_B
c_B K-17 c_A
c_{anchor} c_A, c_B
输入
辅助LLM $\mathcal{M}$
1. 同源块语境检测
“c_A: erratum claim c_B: code binding”
2. 实体指代关系检测
“Tim Cook via K-17: unsupported”
3. 锚点一致性检测
“c_{anchor}: Sam Altman (2019) → conflict”
跨块关系冲突检测
Verdict: remove
$\mathcal{U} = \{c_A, c_B\}$
$\mathcal{C}^* = \text{TOP}_m(\mathcal{C}_{\text{surv}} \setminus \mathcal{U})$
判定结果
Stage 2: LLM判断
“Sam Altman”
防御后的输出